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Stratigraphic implications of palaeomagnetic data from Honduras

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SUMMARY

Palaeomagnetic results from mainly Cretaceous rocks in Honduras delineate a distinctive apparent polar wander path (APWP) which is well suited for magnetostratigraphy. The APWP is based on palaeomagnetic data from palaeontologically dated units in the established stratigraphic column and three radiometrically dated intrusions. The ages and stratigraphic positions of sedimentary units whose stratigraphic assignments have been debated, can be inferred by comparing their palaeomagnetic pole postion with the known APWP. The results demonstrate the usefulness of magnetostratigraphy for resolving stratigraphic problems in the absence of fossils or marker beds, and for areas of discontinuous exposures. For regional interpretations, the most important conclusion is that strata in Honduras that have been assigned to the Todos Santos Formation were not deposited on the same tectonic plate as the Todos Santos Formation of Guatemala and southern Mexico. There is no evidence that the Honduran 'Todos Santos' strata are depositionally related to the true Todos Santos Formation. The name 'Todos Santos Formation' should be abandoned in Honduras and for the entire Chortis Block.

Key words: Honduras, palaeomagnetism, stratigraphy.

1 INTRODUCTION

Geologic field mapping in Honduras faces four main difficulties: poor accessibility, limited outcrop areas, lack of marker beds, and scarcity of diagnostic fossils. Because of logistical problems, only the central part of the republic has been mapped in some detail, and our interpretations pertain only to this area. In this subtropical environment, fresh outcrops are largely restricted to river beds and usually are of limited stratigraphic extent. The paucity of good outcrops combined with the lack of marker bed makes it often impossible to correlate between isolated outcrops.

The direction of the natural remanent magnetization (NRM) of rock samples can, under certain conditions, be used for stratigraphic correlations. Most magnetostratigraphic studies, such as the correlation of sedimentary cores from the ocean basins (e.g. Kennett 1980) or the exact definition of the Cretaceous/Tertiary boundary (e.g. Alvarez *et al.* 1982), are based on the occurrence of reversals of the geomagnetic field and their correlation with the sea-floor magnetic anomaly time-scale. Variations in the direction of the NRM other than reversals form the basis of archaeomagnetism and have been used for the stratigraphic correlation of recent lake sediments (e.g. Creer & Tucholka 1982) as well as older rocks (e.g. Grommé, McKee & Blake 1972).

Another means of using palaeomagnetism for correlation is to compare the magnetic pole position of an unknown rock unit with the pole positions obtained from an established stratigraphic column. This will not only yield a relative age, but also an absolute age with an accuracy which depends on the age resolution of the known apparent polar wander path as well as the associated errors in the palaeomagnetic data (e.g. Porath 1967; Kyle, Ulrich & Gose 1987). Palaeomagnetic results from the Mesozoic section of Honduras delineate an apparent polar wander path which is well suited for this approach. Critical to our work is the fact that some of the rock units had been dated either by fossils or radiometrically. These dated units form the 'calibration points' for the apparent polar wander path, from which the ages and stratigraphic positions of unknown rock units can be inferred by comparison. In this manner we were able to resolve several stratigraphic problems.

2 GENERAL STRATIGRAPHY OF CENTRAL HONDURAS

The stratigraphic section in central Honduras consists of three major parts: a poorly known metamorphic basement of Palaeozoic or older age, a Mesozoic sequence of clastic and carbonate sedimentary rocks, and a Tertiary volcanic cover. The Mesozoic stratigraphy of central Honduras (Fig.

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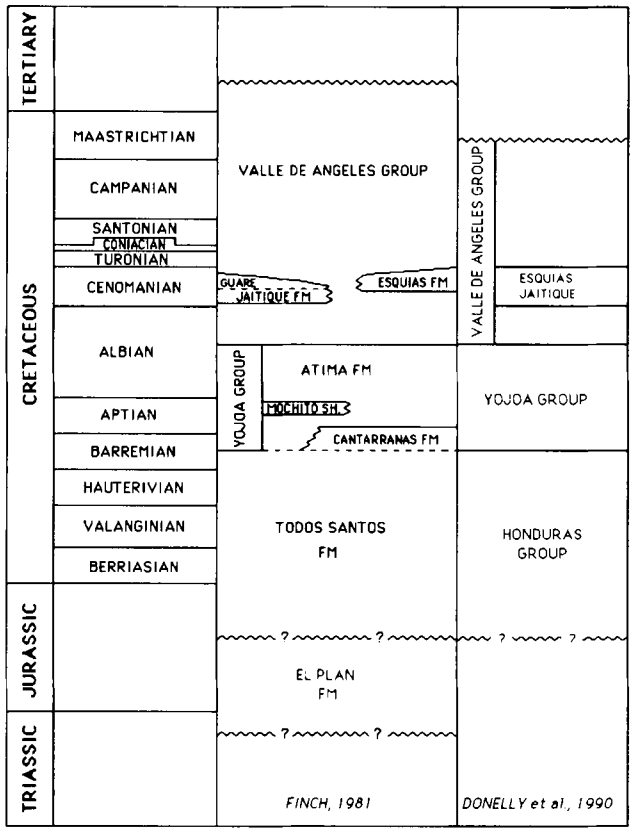


Figure 1. Stratigraphic column for central Honduras after Finch (1981) which was used during the palaeomagnetic sampling and the stratigraphic section proposed by Donelly *et al.* (1990).

1) has been reviewed and revised by Finch (1981). The Jurassic–Early Cretaceous clastic strata have been the subject of further revision by Ritchie & Finch (1985), Finch (1985) and Donelly *et al.* (1990).

The oldest sedimentary unit in Honduras is the Honduras Group (Finch 1985; Ritchie & Finch 1985; Donelly *et al.* 1990). This group includes the El Plan Formation (Carpenter 1954), the 'Agua Fría Formation' (Ritchie & Finch 1985), and unnamed dark-coloured clastic strata that directly overlie metamorphic basement and underlie Cretaceous limestone in various localities throughout Honduras (e.g., the 'basal siliciclastic unit' of Simonson 1977).

The El Plan Formation consists of dark grey shale and siltstone members interbedded with gray to buff sandstone members; minor conglomerate lenses and calcareous beds are present in both members (Carpenter 1954). Mills *et al.* (1967) and Mills & Hugh (1974) extended the use of the name El Plan throughout Honduras; however, the name should probably be restricted to its type area in the San Juancito mining district.

The name 'Agua Fría Formation' (Ritchie & Finch 1985) is being used informally by geologists currently mapping in Honduras due to problems with the definition of the El Plan Formation. 'Agua Fría' strata include thick sections of grey to green shale and siltstone, sandstone, abundant massive quartz pebble conglomerate units and minor coal-bearing strata. Jurassic ferns and cycads have been identified at five localities, and Bajocian to early Bathonian ammonites have been identified at another five localities (Ritchie & Finch 1985; David. G. Taylor, personal communication, May, 1989).

The Honduras Group also includes clastic strata referred to as the 'Todos Santos Formation'. This name was introduced by Sapper (1894) to represent a red bed sequence exposed near the village of Todos Santos in the Altos Cuchumatanes in northwestern Guatemala (site G1 in Fig. 6). It overlies the Permian Chóchal Formation and underlies the Ixcoy or Cobán Formations, two carbonates considered to be time-equivalent with the Atima Formation of Honduras (McBirney 1963; Burkart 1965; Anderson *et al.* 1973). The uppermost member of the Todos Santos Formation is a limestone unit, referred to as the Ventosa Limestone in Guatemala and the San Ricardo Formation in southern Mexico. Based on limited fossil evidence, the age of the Todos Santos is believed to range from Late Jurassic (Oxfordian) to basal Cretaceous (Anderson *et al.* 1973; Guerrero 1975; Richards 1963; Viniegra 1971). Palynological investigations reported by Blair (1981, 1986) on samples from the upper Todos Santos Formation in Chiapas, Mexico, yield an age near the Jurassic–Cretaceous boundary.

Mills *et al.* (1967) applied the same name to red clastic strata in Honduras on the basis of a grossly similar stratigraphic setting, assigning all redbeds underlying Cretaceous limestone to the Todos Santos Formation. Because they did not distinguish between the mainly Aptian–Albian Atima Formation and Cenomanian limestones in the overlying Valle de Angeles Group, later mapped as the Esquías and Jaitique limestones, much of their 'Todos Santos Formation' actually overlies the Atima Formation and belongs to the lower Valle de Angeles red beds as pointed out by Finch (1981).

The Yojoa Group overlies the Honduras Group. The Cantarranas Formation, a shaly limestone, was described by Carpenter (1954), and was redefined as being the basal member of the Yojoa Group by Mills *et al.* (1967). In general, this unit has not been found useful by workers mapping at the 1:50 000 quadrangle scale. The principal member of the Yojoa Group is the Atima Formation, perhaps the most widely recognized rock unit in Honduras. It consists of thick platform carbonate deposits that prograded southward across Honduras in the late Early Cretaceous (mainly Aptian–Albian, but ranging from Barremian to Albian; Mills *et al.* 1967; Lozej 1976; Gallo & van Wagoner 1978; Finch 1981). In the El Mochito Mine near Lake Yojoa, the Atima Formation is unusually thick (at least 1040 m) and is separated into an upper and lower member by the 'Mochito shale' (Lozej 1976). This shale has been correlated by Gallo & van Wagoner (1978) with other shale exposures in central Honduras.

Stratigraphically above the Yojoa Group is the Valle de Angeles Group, a thick red bed sequence divided into unnamed lower and upper clastic members across much of central Honduras by the carbonates of the Jaitique and Esquías Formations (Finch 1981; Horne, Atwood & King 1974; see Fig. 1). The lower red beds generally grade conformably into the Atima Formation (Finch 1972, 1981) and consist of shale, silstone, sandstone, and conglomerate, with the coarser grained units forming a prominent part of the section. Palaeontological dating of the bracketing

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limestones constrains the age of the lower member to Albian–Cenomanian. The upper red beds are mainly fine-grained, and commonly consist of brick-red shales and siltstones. Throughout Honduras the group is unconformably overlain by mid-Tertiary and Miocene volcanic rocks. The Tertiary age reported by Mills *et al.* (1967) for the upper Valle de Angeles has never been documented.

The Plancitos Formation which crops out around the San Juancito intrusive–extrusive complex, located some 25 km northeast of Tegucigalpa, has been described by Carpenter (1954) as consisting of 580 to 1200 m of fairly well-bedded, medium- to coarse-grained andesitic 'grits', interbedded with poorly sorted, fine-grained sandstone, siltstone, and shale. The grits contain fragments of graywacke, silty shale, quartzite, limestone, and schist in addition to the igneous detritus, and grade into large lenses of boulder conglomerate. These sediments are of limited areal extent and were derived from the nearby San Juancito complex. Stratigraphically, the Plancitos Formation has been placed below and above the Cantarranas Formation, within the Valle de Angeles Group, and even a Tertiary age has been advocated (Carpenter 1954; Mills *et al.* 1967; Williams & McBirney 1969; Wilson 1974).

The Mesozoic sedimentary section has been intruded by granitic rocks at numerous localities and also contains some extrusive units. It is unconformably overlain by Tertiary volcanic rocks of the Matagalpa Formation and Padre Miguel Group. The oldest radiometric age thus far obtained from these rocks is about 30 Myr (K–Ar mineral ages: McDowell, personal communication, 1987).

3 PALAEOMAGNETIC DATABASE

The palaeomagnetic sampling programme included sampling sites specifically selected to test the stratigraphic work of Finch. Samples were collected from all Mesozoic sedimentary formations identified at the time and also from three radiometrically dated intrusions. A detailed discussion of the data and their tectonic interpretation is given in Gose (1985a).

Table 1 summarizes the virtual geomagnetic pole positions (VGPs) for published palaeomagnetic results from Honduras (Gose 1985a). These entries differ from the original data only in that results from some closely related sites have been combined. The data are arranged in age sequence based on palaeontological or radiometric determinations, or physical, stratigraphic relationships.

The sampling sites are distributed throughout west-central Honduras (Fig. 2), the same area to which the stratigraphic information of Fig. 1 applies. The region east of the Guayape fault is poorly accessable and the geological information is largely based on air photography. The tectonic interpretation of the palaeomagnetic data directly applies, of course, only to the sampling area which is the centre of the continental Chortis Block. The northern limit of this block is defined by the Polochic and Motagua faults in Guatemala (Fig. 2); its southern limit has been placed, somewhat arbitrarily, at 12.5°N because of the lack of geological information (Dengo & Bohnenberger 1969). The original definition referred only to the land area between these boundaries. The Middle America Trench forms the present western boundary, but it seems likely that the eastern edge of the Nicaraguan Depression (east of Lake Nicaragua) delineates the original boundary because it separates the Cretaceous–Tertiary sedimentary basins of the Pacific coast from the Tertiary volcanic plateau of the interior highland. The Caribbean edge of the Chortis Block

Table 1. Palaeomagnetic pole positions (VGPs) of sampling sites in Honduras arranged in stratigraphic sequence. N = number of samples used in the analysis; k = precision parameter (Fisher 1953), LAT and LONG are the coordinates of the pole positions; dp and dm represent the axes of the ellipse of 95 per cent confidence. Data are structurally corrected. The ages of the igneous rocks are: Depilto batholith: 145 ± 15 Myr; San Francisco Adamelite 114 ± 1.8 Myr; Minas de Oro Granodiorite: 59.3 ± 0.9 Myr (based on Gose 1985a).

FORMATION	N	k	LAT	LONG	dp	dm
MINAS DE ORO DIORITE, M1+M2	18	52.8	48.9 N	190.6 E	3.0	5.3
VALLE DE ANGELES , V11-V13	51	32.6	18.0 N	192.1 E	2.0	3.8
ESQUIAS FM, E1+E2	35	202.7	16.0 N	198.4 E	1.1	1.9
JAITIQUE FM, J3+J4	25	32.3	6.2 N	201.9 E	3.6	6.1
JAITIQUE FM, J1+J2	20	22.9	4.1 N	200.8 E	4.6	8.0
VALLE DE ANGELES, V7-V10	50	45.4	2.3 S	194.5 E	1.7	3.2
VALLE DE ANGELES, V4-V6	39	38.6	39.7 S	217.6 E	2.3	4.1
VALLE DE ANGELES, V1-V3	33	70.0	53.6 S	258.4 E	2.1	3.5
ATIMA FM, A9+A10	26	32.1	29.1 S	223.3 E	4.0	6.4
ATIMA FM, A8	19	38.9	15.4 S	217.4 E	4.5	7.0
ATIMA FM, A6+A7	18	63.1	8.0 S	209.8 E	3.2	5.3
ATIMA FM, A5	11	63.9	2.8 S	199.1 E	3.5	6.4
SAN IGNACIO ADAMELLITE, SI1	8	70.7	1.5 N	201.2 E	4.3	7.6
ATIMA FM A2+A3	57	55.7	29.7N	197.6 E	2.0	3.4
ATIMA FM, A1+A4	38	37.4	14.9N	197.7 E	2.0	3.4
CANTARRANAS FM, C1	20	30.2	16.7 N	196.7 E	3.9	6.8
"TODOS SANTOS" FM, TS1+TS2	71	42.1	26.0 N	192.2 E	1.6	2.9
DIPILTO GRANODIORITE, DP1	5	51.9	31.2 N	191.1 E	6.2	11.6

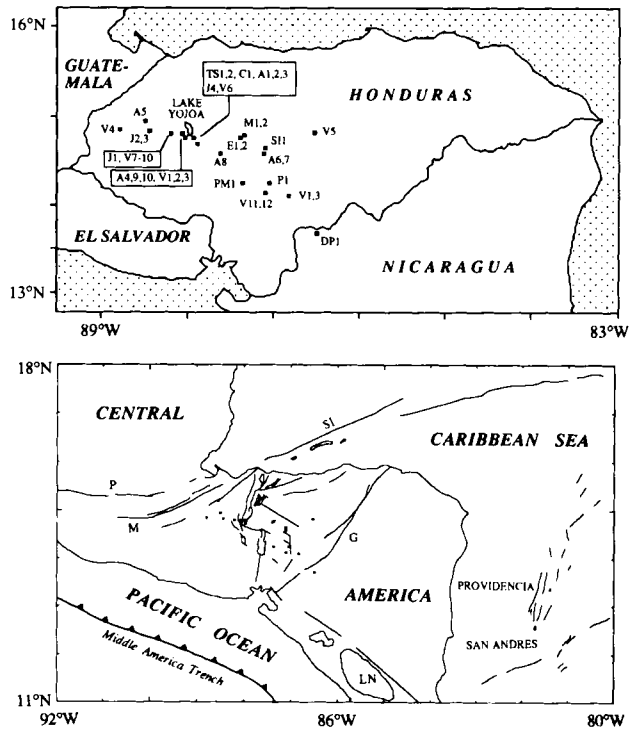


Figure 2. Distribution of sampling areas and simplified tectonic map of the Chortis Block (modified from Case & Holcombe 1980). P = Polochic fault; M = Motagua fault; SI = Swan Islands transform fault; G = Guayape fault; LN = Lake Nicaragua.

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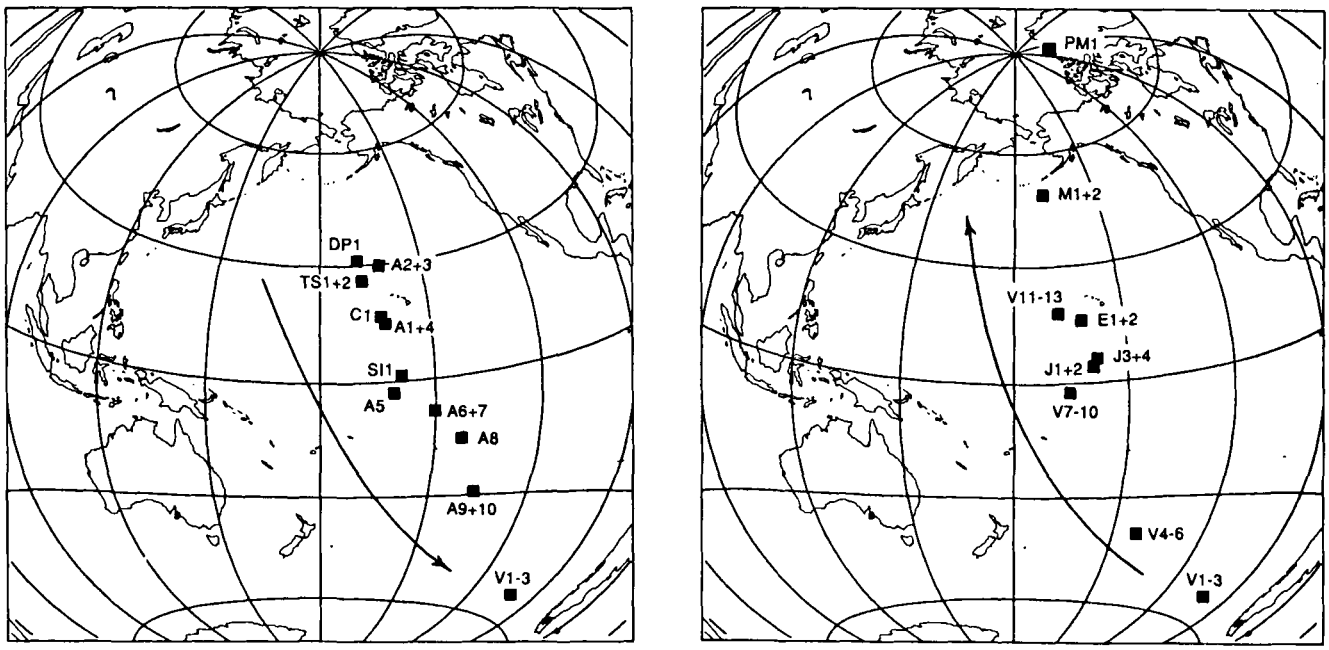


Figure 3. Virtual geomagnetic pole positions coded to Table 1. Large arrows indicate age progression. Pole PM1 is a new data point listed in Table 2.

has not been identified. Seismic activity and the presence of a series of normal faults suggest a N15°E striking line through the San Andrés and Providenica islands as a possible eastern limit (Gose 1985b). The northern marine limit is the Swan Islands transform fault. The Mesozoic rocks of the Chortis Block have been folded and faulted in the Cainozoic, but the deformation is relatively minor (e.g. Elvir 1974; Muehlberger 1976; Case & Holcombe 1980; Mann & Burke 1984).

The virtual geomagnetic pole positions (VGPs) delineate a distinctive apparent polar wander path (Fig. 3). These data have been interpreted by Gose (1985a) as reflecting large tectonic rotations for the Chortis Block during the Cretaceous. The data also suggest that the Chortis Block sutured onto southern Mexico/Guatemala (the Maya Block) at the closing of the Cretaceous Period, an interpretation which is strongly supported by other geological data (Gose 1985a; Donelly *et al.* 1990).

In order to use these data for stratigraphic interpretations it is important to establish that the apparent polar wander path is not the result of local tectonics. Apparent tectonic rotations due to incorrect tilt corrections (MacDonald 1980) cannot produce the observed polar wander path because the sampled strata dip between 12° and 45° and the maximum angle of apparent rotation cannot exceed the dip angle. Thus the observed apparent polar wander path cannot be the result of mismeasured dips. It is equally unlikely that local rotations about a vertical axis play a significant role. For example, site J3 lies to the west of the Honduran Depression, the series of roughly north–south striking grabens in west-central Honduras between longitude 87° and 88°W, and site J4 is located within the Depression (Fig. 2). Their pole positions are statistically indistinguishable. Yet the pole position from site V6, which lies on the same structural block as site J4, does imply a large rotation. Additional examples are given in Gose (1985a). A fold test could be applied to most sites and demonstrates that the magnetization is at least pre-Lower Tertiary (table 2 in Gose 1985a). It is also important to point out that the palaeomagnetic data form an internally consistent data set in which the time sequence of pole positions conforms to all stratigraphic constraints.

The most dramatic change in the pole posistions is reflected in the variations of the latitude of the VGPs (this latitude is not to be confused with the palaeolatitude of the sites; the Chortis Block was very close to its present latitude throughout the Cretaceous). For the purpose of correlation, it is convenient to look at the VGP latitude of the sampling sites as a function of age (Fig. 4). For each data point, we plotted, as a vertical bar, the age range of the strata as reported in the literature (Mills *et al.* 1967; Lorzej 1976; Gallo & van Wagoner 1978; Finch 1981; Gose 1985a). The VGP latitude with its error limit is shown by the horizontal bar. The central part of the figure shows the geomagnetic reversal time-scale and the geological stages after Palmer (1983). To the right is our revised stratigraphic column for central Honduras.

Superposed on the palaeomagnetic data is the generalized trend of these data points (thin 'lazy V' line) which is constrained by those palaeomagnetic data for which we have the best age control. These points are identified in the figure by solid circles. The stratigraphic position of rock units of unknown or poorly constrained age may be determined by plotting their VGP latitude on this reference curve. The accuracy of the stratigraphic assignment will depend on the error limit of the new palaeomagnetic data and the age uncertainty and palaeomagnetic error of the reference data.

4 STRATIGRAPHIC INTERPRETATIONS

4.1 'Todos Santos' Formation

Near Pito Solo at the south shore of Lake Yojoa, we sampled strata assigned to the 'Todos Santos' Formation

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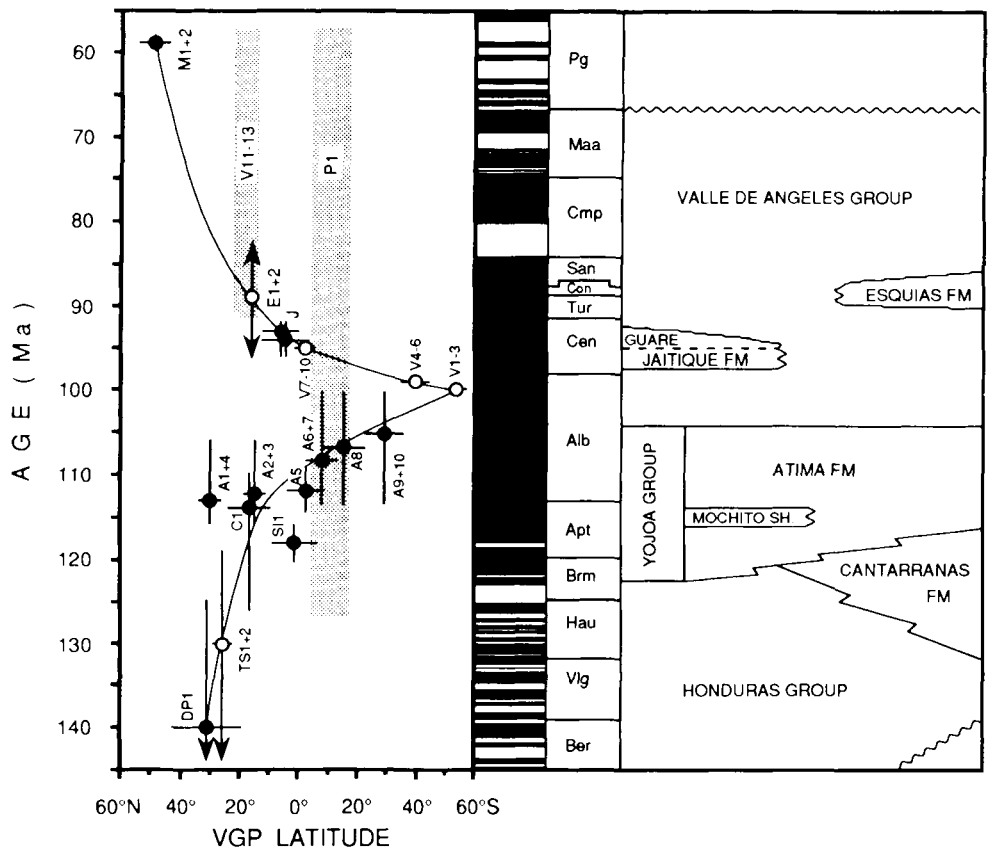


Figure 4. Latitude of virtual geomagnetic pole positions (VGPs) from sampling sites in Honduras versus time. Vertical bars indicate the age range of the site, horizontal bars the uncertainty in latitude. We use *dm*, the major axis of the VGP error ellipse (95 per cent confidence level) because *dm* is equal to or greater than the actual latitudinal error. Solid circles indicate those data for which we have the best age control. The age of sites V1–6 is determined by physical stratigraphy only. The thin 'lazy V' line indicates the general trend of the palaeomagnetic data. The shaded rectangles delineate the age range and latitudinal error for poles P1 and V11–13. Magnetic reversal sequence and stage assignment after Palmer (1983). To the right is our revised stratigraphic column.

(Mills *et al.* 1967; Curran 1981a). From these samples, palynologists with Texaco Trinidad, Inc. extracted the following palynomorphs: *Eucommiidites troedssoni*; *Cicatricosisporites dorogenesis*; *Classopolis classoides*; *Cyathdites minor*; *Circulina parva*; *Ginkocyadophytus cf. nitidus* (T. L. Fyock, written communication, 1978). This association indicates an Early Cretaceous age which corresponds reasonably well to the age of the upper Todos Santos Formation in Mexico. At Pito Solo, the overlying Cantarranas Formation has been dated by Mills *et al.* (1967) as early Albian. The palaeomagnetic results for this and a nearby second site are listed in Table 2 (from Gose 1985a).

In Guatemala, we sampled the Todos Santos Formation at two locations. In the Altos Cuchumatanes Mountains near the village of Todos Santos, we took 18 samples over a stratigraphic interval of 27 m in the upper portion close to the contact with the Ventosa Limestone (site G1, see Fig. 6). The second locality (G2) is some 47 km south of Cobán, where the Todos Santos is in fault contact with the Permian Chóchal Formation. 10 samples, spanning 50 m stratigraphically, were collected from what Walper (1960) considered to represent the basal part of the Todos Santos Formation. The NRM intensities of most samples are around 10^{-6} A m^2 kg^{-1}. All samples were thermally demagnetized in seven steps up to 665 °C and the characteristic directions were calculated using the directions observed from 500° or 600° to 665 °C. The directions of the stable magnetization for both sites and representative vector diagrams are shown in Fig. 5. The statistical parameters are listed in Table 2.

The mean direction, corrected for bedding tilt, of the four normally magnetized samples is indistinguishable from the mean of the reversely magnetized samples at the 1 per cent significance level. At each locality, the bedding attitudes vary little so that no local fold test can be applied. However, a regional fold test can be carried out by combining the two data sets. In this case, the data cluster better at the 99 per cent confidence level after structural correction, implying that the magnetization predates the folding. Thus there is strong evidence that the primary magnetization has been successfully isolated.

Figure 6 compares the new data from Guatemala with our previous data from Honduras. Also shown in this figure is the pole position M1 from the San Ricardo Formation, the uppermost member of the Todos Santos Formation of southern Mexico (Guerrero 1975; Guerrero, Herrero-Bervera & Helsley 1990). Our results from Guatemala agree with the approximately time-equivalent data from Mexico, but differ markedly from the Honduran 'Todos Santos' results. This is really not surprising in view of the position of the sites on two different tectonic plates. The Chortis Block was not part of the North American plate nor of a Caribbean block throughout the Cretaceous (Gose & Swartz

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Table 2. Statistical parameters of published palaeomagnetic analyses for the Todos Santos Formation (TS1, 2 from Gose 1985a; M1 from Guerrero 1975) and new data reported here. For the new data, the first line lists the data after structural correction, the second line before tilt correction. SITE = east longitude and north latitude of sampling site; TREAT = statistics were obtained by using the specified temperature (°C) or by analysis of orthogonal vector projections; N = number of samples included/rejected from analysis; R = resultant vector; DECL, INCL = mean declination and inclination; k = Fisher's precision parameter; α_{95} = radius of 95 per cent circle of confidence; LAT, LONG = north latitude and east longitude of pole position; dp/dm = semi-axes of 95 per cent confidence ellipse.

FORMATION	SITE	TREAT	N	R	DECL	INCL	k	α 95	LAT	LONG	dp/dm
"TODOS SANTOS FM" HONDURAS, TS1	272.0/14.8	TD 500	44/5	42.9	294.1	31.0	38.9	3.5	26.9	193.6	2.2/3.9
"TODOS SANTOS FM" HONDURAS, TS2	272.0/14.8	TD 500	27/7	26.5	292.5	25.5	54.4	3.8	24.8	190.2	2.2/4.1
TODOS SANTOS FM MEXICO, M1	266.7/16.8	TD 630	89/12	85.5	340.3	19.9	25.4	3.0	69.8	160.0	1.7/3.2
TODOS SANTOS FM GUATEMALA, G1	268.5/15.5	ORTHO	15/3	14.4	156.1	-14.1	22.1	8.3	65.1	161.5	4.4/8.5
				14.4	139.8	-32.8	23.7	8.0			
TODOS SANTOS FM GUATEMALA, G2	269.8/15.2	ORTHO	7/3	6.9	324.5	7.1	45.8	9.0	53.1	164.9	4.6/9.1
				6.9	324.2	-7.3	42.4	9.4			
PLANCITOS FM HONDURAS, P1	273.0/14.3	ORTHO	14/2	13.7	252.9	35.7	50.0	5.7	-10.7	206.7	3.8/6.6
				13.7	248.8	16.6	48.3	5.8			
PADRE MIGUEL GROUP HONDURAS, PM1	272.6/14.3	ORTHO	17/1	16.9	1.7	38.8	320.3	2.0	82.2	284.0	1.4/2.4
				16.9	346.2	58.9	290.2	2.1			

1977; Gose 1985a). The lack of physical contiguity between the true Todos Santos and the Honduran 'Todos Santos' along with the independent tectonic history of the Chortis Block strongly implies that these units were not deposited in the same depositional basin. We concur with Gose (1977) that the name Todos Santos should be abandoned in Honduras and the rest of the Chortis Block. Ritchie & Finch (1985) and Donelly *et al.* (1990) place the strata, formerly mapped as 'Todos Santos Formation', as an unnamed unit into the Honduras Group.

4.2 PLANCITOS FORMATION

At a fresh roadcut, we collected 16 samples of the Plancitos Formation over a stratigraphic interval of 12 m. All samples were demagnetized in nine steps by alternating field, followed by thermal demagnetization to 565 °C. The NRM intensities of the samples ranged from 3×10^{-7} to 2×10^{-5} A m^2 kg^{-1}. A present field overprint was removed after thermal demagnetization to 300° to 400 °C and the characteristic direction was calculated typically over the

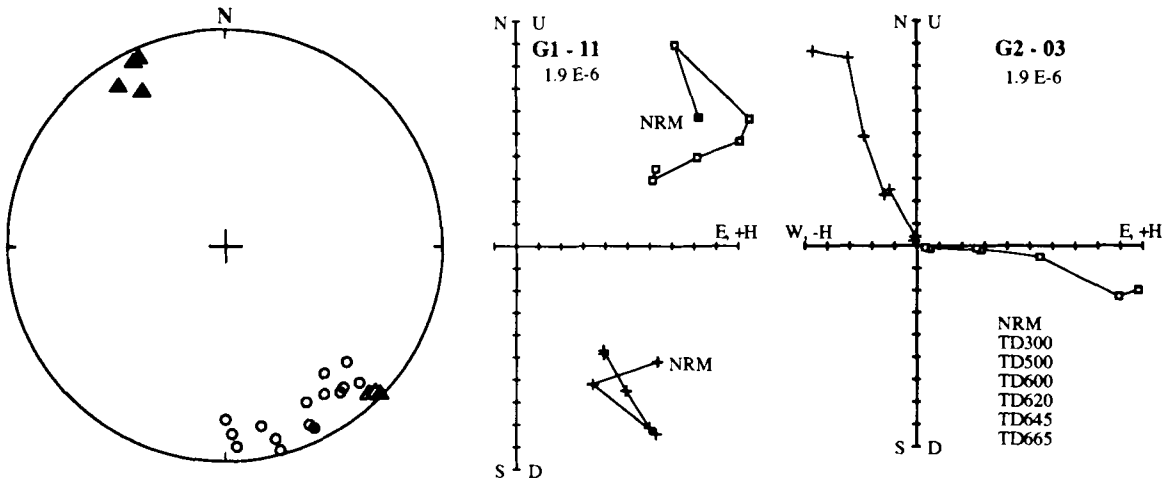
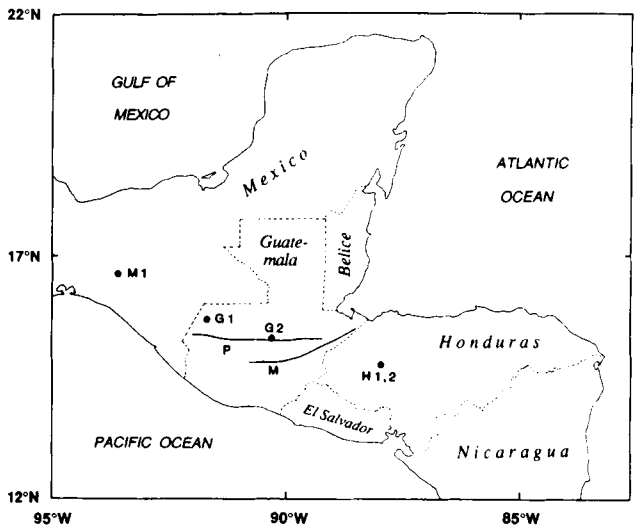


Figure 5. Schmidt equal area projection of the characteristic direction of magnetization for samples from the Todos Santos Formation in Guatemala (site G1 = circles, site G2 = triangles). Solid symbols are in the lower hemisphere, open symbols in the upper. To the right are two examples of orthogonal vector projections. Crosses are in the N/S–E/W plane and squares in the up/down–horizontal plane. The number column indicates the demagnetization temperatures in °C. The intensities of magnetization are given in A m^2 kg^{-1}. All data are corrected for bedding tilt. Same convention applies to all similar figures.

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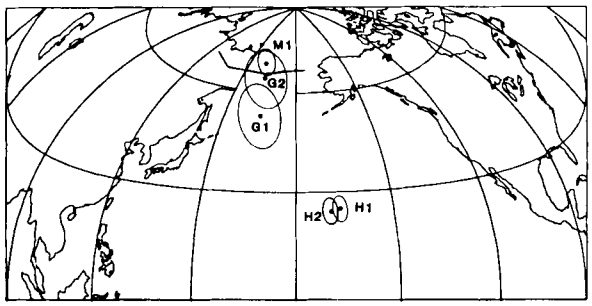


Figure 6. Top: location map of sampling sites of the Todos Santos Formation in Mexico (M1, from Guerrero 1975) and Guatemala (G1, type section, and G2) and strata in Honduras (H1 and H2) previously mapped as Todos Santos. M = Motagua Fault, P = Polochic Fault. Bottom: palaeomagnetic pole positions from the Honduran sites are distinctly different from the Todos Santos sites from north of the Polochic Fault. Line crossing the error ellipses (95 per cent confidence) shows the North American apparent polar wander path from 120 Myr (at the eastern end) to 160 Myr (after Irving & Irving 1982).

demagnetization range from 400° to 565 °C. Fig. 7 shows directions of the stable magnetization and an example of a vector component diagram. The statistical parameters are listed in Table 2. The latitude of the palaeomagnetic pole position for the Plancitos Formation clearly precludes a Tertiary age and instead implies a 'middle' Cretaceous age (Fig. 4). This latitude permits the Plancitos to be correlated with either of two units: the Aptian–Albian Yojoa Group or the Albian–Cenomanian lower Valle de Angeles red beds. Our own field work favours the older age assignment. There is also a tenuous indication in the palaeomagnetic data which favours the older age. The single reversely magnetized sample could correspond to one of several short reversed polarity events found in the Late Aptian–Early Albian part of the Cretaceous Normal Polarity Superchron (Hailwood, Hamilton & Morgan 1980; Tarduno 1989). No reversed polarity horizons have been reported in the younger part of this chron.

4.3 Valle de Angeles Group: Guare member of the Jaitique Formation

The Guare member is a distinctive, thin-bedded, algal (?) laminated limestone exposed in the Santa Bárbara–Lake Yojoa region of central Honduras. The name was introduced by Mills *et al.* (1967) as the uppermost unit of the Yojoa Group, overlying the Atima Formation. Indeed, the Guare everywhere directly overlies a thick-bedded limestone. However, quadrangle mapping by Finch (1972, 1979, 1985) and Curran (1980, 1981a,b) demonstrated that the Guare overlies the Cenomanian Jatique limestone rather than the Atima Formation. Finch (1981) redefined the Guare as the upper member of the Jaitique Formation. By themselves, the palaeomagnetic results from two sites in the Guare (J3 and J4 in Table 1 and Fig. 3) would permit two interpretations: (1) the Guare is part of the lower Atima Formation; or (2) the Guare is the uppermost member of the Jaitique Formation. The first interpretation is not tenable because the Guare is overlain by gypsum and red beds, not limestones. The second interpretation, on the

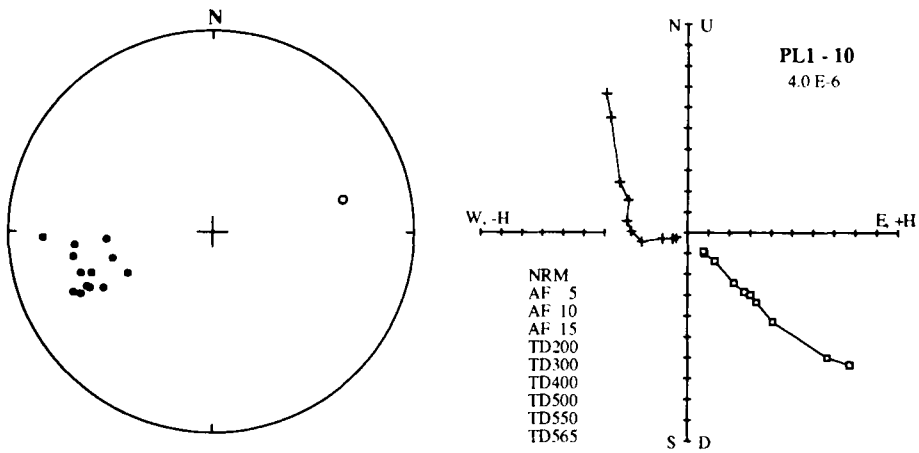


Figure 7. Characteristic directions of magnetization for samples from the Plancitos Formation (structurally corrected). Note that one sample is reversely magnetized. Vector diagram shows the removal of a present field overprint during alternating field demagnetization (values in mT) and thermal demagnetization.

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other hand, corraborates all recent stratigraphic mapping in this region.

4.4 Valle de Angeles Group: Jaitique and Esquías Formations

Two limestones have been mapped in central Honduras, the Esquías Formation (Horne *et al.* 1974) and the Jaitique Formation (Finch 1981). Microfossils clearly establish a Cenomanian age for the Jaitique Formation, but the generally quoted Cenomanian age for the Esquías Formation is only poorly constrained. Palaeontological identifications on Esquías samples have yielded ages ranging from late Albian to early Tertiary, and the most complete collection can only establish an 'apparently middle' Cretaceous age (Horne *et al.* 1974). Indeed, K. Young, who identified most of these fossils, favours a Turonian age (K. Young, personal communication, 1988). It has not been possible thus far to establish by field mapping whether the two units were contiguous when deposited, were separated but time-equivalent deposits, or were spatially and temporally separate units. Whether or not the use of both formation names is justified has been a subject of debate amongst some workers in Honduran stratigraphy.

The Jaitique Formation, including the Guare member, was sampled at four localities and the Esquías Formation at two localities. At each locality, the samples were drilled over stratigraphic intervals ranging from 5 to 27 m. In order to compare the data at the site level, we define a site as consisting of three or four neighbouring samples typically spanning about 1 m stratigraphically. This approach yields 13 sites for the Jaitique Formation and 11 sites for the Esquías Formation. The mean direction of the stable magnetization for the 13 Jaitique sites is declination = 270.0°, inclination = 36.5°, and the 95 per cent circle of confidence is $\alpha_{95} = 6.7°$. The corresponding values for the 11 Esquías sites are $D = 282.3°$, $I = 34.5°$, $\alpha_{95} = 2.3°$. The two directions are statistically different at the 99 per cent significance level and the Jaitique and Esquías Formations are, therefore, regarded as not time-equivalent. The pole position for the Esquías (16.0°N, 198.4°E) lies 11° to the north of the Jaitique pole (5.0°N, 201.6°E) which implies that the Esquías Formation is younger than the Cenomanian Jaitique Formation by a few million years based on the overall trend of the Honduran apparent polar wander path (Fig. 4; see also fig. 16 in Gose 1985a). The implied age difference plus the lithologic difference (the distinctive Guare Member topping the Jaitique Formation is not found in the Esquías Formation) support the usage of separate formation names.

4.5 Valle de Angeles Group: red beds

The pole positions for the three sites in the upper Valle de Angeles red beds (pole V11, 12, 13), one of which is within 10 m of the upper angular contact with volcanic strata, fall very close to the pole positions for the Esquías Formation (E1, E2 in Fig. 4 and Table 1). They are distinctively south of the pole from the Minas de Oro stock which is dated at 59 Myr (see Gose 1985a). Thus, at least where sampled, the Valle de Angeles Group is restricted to the Cretaceous Period.

Two of these three sites (V11 and V12), separated horizontally by some hundred metres, represent a 25 m section which is reversely magnetized. Gose & Swartz (1977) suggested that this reversal may correspond to the first reversal after the long normal mid-Cretaceous chron and assigned a tentative Coniacian age to these sites using the reversal time-scale of Larson & Pitman (1972). Based on more recent time-scales (Palmer 1983; Harland *et al.* 1989), the age of this reversal is early Campanian. In principle, this reversal could be any reversal younger than the long normal interval, but the closeness of the pole position to the Esquías pole and its angular distance from the Minas de Oro pole makes the Campanian assignment more attractive.

Samples from these two sites were subsequently analysed by palynologists from Texaco Trinidad and yielded these palynomorphs: *Classopolis classoides*, *Ephedripites sp.*, *Cirulina parva*, and *Classites sp.* (T. L. Fyock, written communication, 1978). Because only a few, badly preserved pollens were observed, their age can not be specified more precisely than Barremian to Turonian. The poor preservation state of these pollens suggests reworking. Thus their age is only a maximum age for these red beds, and the palaeomagnetically inferred age seems reasonable.

In spite of some overall sedimentological differences between the upper and lower Valle de Angeles red beds (Finch 1981), it is often difficult to distinguish between these units where neither the Jaitique nor Esquías Formation is present. Magnetically, however, the two members are rather distinct. The lower red beds yield palaeomagnetic poles (V1 to V10) that lie south of the Jaitique and Esquías poles; the upper red beds give poles (V11 to V13) that fall to the north. To test the validity of this generalization, we sampled two outcrops, one in western Honduras near Santa Rosa de Copán (pole V4, Fig. 2), and one some 270 km to the east near Salamá (V5). These outcrops had not been mapped and their stratigraphic positions were unknown. At both sites, the strata are rather coarse-grained, a feature which is more typical of the lower redbeds (Finch 1972, 1981), so our preliminary interpretation was to assign these outcrops to the lower member. Their pole positions clearly fall in the range of poles from known lower member sites (Fig. 4), confirming our provisional interpretation.

5 CONCLUSIONS

For regional interpretations the most important conclusion is that the clastic strata in Honduras, formerly assigned to the Todos Santos Formation, were deposited on a different tectonic plate from the true Todos Santos Formation of Guatemala and southern Mexico; the name 'Todos Santos Formation' should, therefore, be abandoned in Honduras and the rest of the Chortis Block. These strata are now an unnamed member of the Honduras Group. The following other findings are mainly of local significance but do demonstrate the power of magnetostratigraphy. (1) The non-fossiliferous exposures of red beds belonging to the Valle de Angeles Group can be assigned to either the upper or lower member based on their direction of magnetization. (2) The upper Valle de Angeles beds, at least where sampled for this study, are of Late Cretaceous, not Tertiary age as reported in earlier literature. (3) The Guare limestone is the upper member of the Jaitique Formation as defined by Finch (1981). (4) The Jaitique Formation is older than the Esquías Formation, and the Esquías limestone seems to be younger than the generally quoted Cenomanian

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age. (5) The Plancitos Formation is of middle Cretaceous age.

The stratigraphic column in Fig. 4 combines the results of recent field work by Ritchie & Finch (1985, 1991) with the results of this study. We view this revised column as our best working model, but do expect that future work may further modify or refine this column. It is important to date the Esquías Formation palaeontologically and to analyse new samples from the upper Valle de Angeles for their fossil content. The 'Agua Fría Formation', unknown at the time of the palaeomagnetic sampling, clearly should be studied palaeomagnetically. Because of the success of magnetostratigraphy, sampling of the poorly mapped eastern part of Honduras can now be attempted.

6 AN ADDITIONAL DATA POINT

The main goal of the palaeomagnetic sampling programme was to obtain data from Cretaceous rocks. After the initial results (Gose & Swartz 1977) indicated large rotations it was deemed worthwhile to collect a much younger rock unit. Projecting the polar wander path towards the present suggests that such a young rock should reveal little or no rotation. The Padre Miguel Group in the Zambrano area of central Honduras is a sequence of gently dipping (24°) ignimbrites. The particular ignimbrite which we sampled has not been dated, but five nearby units have yielded K–Ar ages between 10 and 14 Myr (analyses by F. McDowell quoted in Dupré 1970).

We collected 17 samples distributed throughout a 20 m thick ignimbrite and one sample from the overlying ignimbrite. The samples are strongly magnetized ($\sim 10^{-3}$ A m^2 kg^{-1}). With a few exceptions, the *in situ* NRM directions group in a direction well removed from the present dipole field direction (Fig. 8). The single reversely magnetized sample (#18) comes from the younger flow. All samples were stepwise demagnetized to 600 °C and the characteristic directions were calculated using the demagnetization steps from 350° or 500° to 600 °C. The vector diagrams in Fig. 8 attest to the high magnetic stability. The sample closest to the upper contact (#17) has been partially remagnetized by the reversely magnetized overlying flow. The fact that this sample retained its intermediate direction demonstrates magnetic stability of the ignimbrite on the geological time-scale (contact test). For the purpose of calculating the mean, this sample will be deleted. Because the one sample from the younger ignimbrite is magnetized exactly antipodal to the other samples, it is included in the statistics (Table 2). As expected, the palaeomagnetic pole position after structural correction lies close to the geographic pole (Fig. 3). Realizing that the direction of magnetization of this fast cooling rock unit is affected by secular variation, this data point suggests that, by Middle Miocene time, the sampling site (and the Chortis Block?) was close to its present position relative to North America.

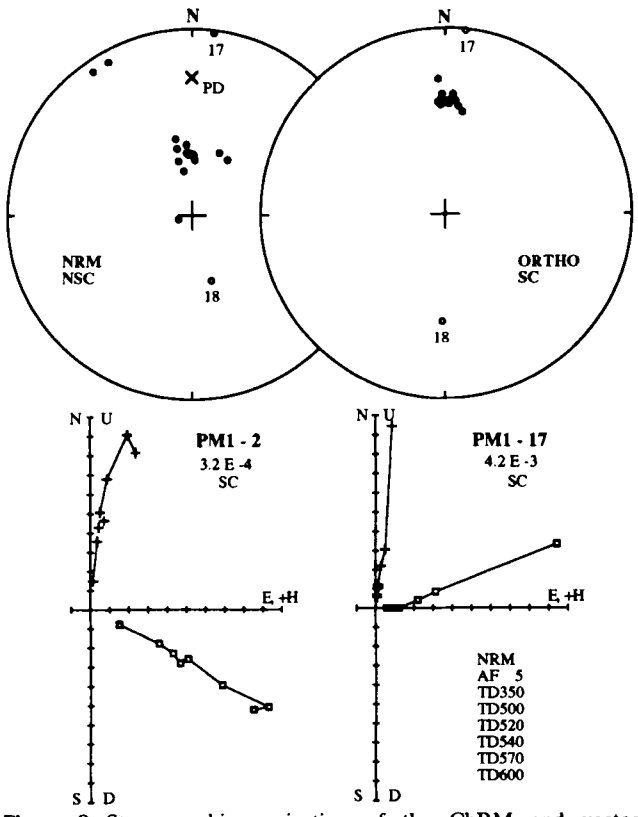


Figure 8. Stereographic projection of the ChRM and vector diagrams for the Padre Miguel ignimbrite. Most *in situ* NRM directions cluster in a direction well removed from the present dipole filed (PD). Sample 17 has been remagnetized by the overlying, reversely magnetized ignimbrite (sample 18).

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